

**GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY****(AN AUTONOMOUS INSTITUTION)****(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)****(Accredited by NAAC with "A" Grade, NBA (EEE, ECE & ME) & ISO 9001:2008 Certified Institution)****QUESTION BANK (DESCRIPTIVE)****Subject Name: Big Data Analytics****Subject Code: 23A3214T****Course & Branch: B.Tech C S (D S)****Year & Semester: IV B.Tech I Semester****Regulation: RG23****Unit 1**

S. No	2 Marks Questions (Short)	[BT Level] [CO] [Marks]
1.	Define Big Data?	L1, CO1, 2M
2.	Define structured, semi-structured, and unstructured data?	L1, CO1, 2M
3.	Define Web Analytics and mention any two industry applications of Big Data?	L1, CO1, 2M
4.	What is Big Data in Marketing? Mention its role in Advertising?	L1, CO1, 2M
5.	Define Fraud Analytics, and Credit Risk Management?	L1, CO1, 2M
6.	What is Algorithmic Trading?	L1, CO1, 2M
7	Name any four Big Data technologies?	L1, CO2, 2M
8.	Define Hadoop. Mention any two advantages of Hadoop.	L1, CO2, 2M
9	What are Open Source Technologies?	L1, CO2, 2M
10	Define Cloud Computing & Firewall analytics?	L1, CO2, 2M

S. No	Descriptive Questions (Long)	[BT Level] [CO] [Marks]
1	Explain the concept of Big Data and discuss the need for Big Data in modern organizations.	L2, CO1, 10M
2	Describe the convergence of key trends that led to the emergence of Big Data. Explain its (5Vs) of Big Data with suitable examples.	L2, CO1, 10M
3	Differentiate between structured, semi-structured, and unstructured data with suitable examples.	L2, CO1, 10M
4	Explain various industry examples and applications of Big Data. Discuss the role of Big Data in different business sectors.	L2, CO1, 10M
5	Explain the concept of Web Analytics. Discuss how Big Data is used in Marketing and Advertising with suitable examples.	L2, CO1, 10M
6	Explain the concept of Fraud Analytics. Discuss the role of Big Data in Fraud Detection with suitable examples.	L2, CO2, 10M
7	Discuss the applications of Big Data in Risk Management and Credit Risk Management. Explain their significance in financial institutions.	L2, CO2, 10M
8	Explain the concept of Algorithmic Trading. Discuss the role of Big Data in automated trading systems with suitable examples.	L2, CO2, 10M

9	Discuss the applications of Big Data in Healthcare and Medicine. Explain how Big Data improves patient care, diagnosis, and medical research.	L2,CO2,10M
10	Describe various Big Data Technologies. Explain the Hadoop framework and its Architecture with Suitable Examples?	L2,CO2,10M
11	Write a Short Note on Mobile Business Intelligence and Crowd Sourcing Analytics in BDA with suitable examples.	L2,CO2,10M
12	Discuss the importance of Open Source Technologies. Explain the relationship between Cloud Computing and Big Data.	L2,CO2,10M
13	Write a Short Note on Inter-firewall Analytics and Trans-firewall Analytics in BDA with Suitable Examples?	L2,CO2,10M

Unit-2

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define NoSQL	L1,CO3,2M
2.	What are aggregate data models?	L1,CO3,2M
3.	Define Key-Value data model and Document data model.	L1,CO3,2M
4.	What are relationships in NoSQL databases?	L1,CO3,2M
5.	Define Schema-less Database and Materialized View.	L1,CO3,2M
6.	What is a distribution model?	L1,CO3,2M
7.	Differentiate between Master-Slave Replication and Peer-to-Peer Replication.	L1,CO3,2M
8.	What is sharding and replication?	L1,CO3,2M
9	What is meant by relaxing consistency and version stamps?	L1,CO3,2M
10	Write the steps involved in creating a table, loading data, and reading data in Cassandra.	L1,CO3,2M

S.No	Descriptive Questions(Long)	[BT Level] [CO][Marks]
1	Explain the concept of NoSQL databases. Discuss the advantages and limitations of NoSQL databases over traditional relational databases.	L2,CO3,10M
2	Explain the Aggregate Data Model?&Discuss the concept of its Aggregates with suitable examples.	L2,CO3,10M
3	Write a short Note on Key-Value Data Model and Document Data Model with Suitable Examples?Explain it?	L2,CO3,10M
4	Explain the relationships in NoSQL databases& Graph Databases with a neat sketch?	L3,CO3,10M
5	Write a short note on Schema-less Databases & Materialized Views in NoSQL databases with suitable Examples?	L2,CO3,10M
6	Discuss the Distribution Models used in NoSQL databases. Explain the concept of Sharding and its significance in distributed database systems.	L2,CO3,10M
7	Explain the Master-Slave Replication model. Compare it with the Peer-to-Peer Replication model with suitable diagrams.	L2,CO3,10M
8	Discuss the concepts of Sharding and Replication. Explain how they improve scalability, availability, and fault tolerance in distributed databases.	L2,CO3,10M
9	Explain the concept of Consistency in distributed databases. Discuss Relaxing Consistency and Version Stamps with suitable examples.	L2,CO3,10M
10	Explain the architecture of Apache Cassandra. Discuss its role as a distributed NoSQL database.	L2,CO3,10M
11	Explain the process of creating tables, loading data, and reading data	L2,CO3,10M

	in Cassandra with suitable CQL commands and its examples.	
12	Write a short Note on Apache Cassandra by explaining data storage, replication, consistency, query processing, and practical applications.	L2,CO3,10M

Unit-3

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define data formats.	L1,CO4,2M
2.	What is Hadoop?	L1,CO4,2M
3.	Define HDFS	L1,CO4,2M
4.	What is fault tolerance?	L1,CO4,2M
5.	Define High Availability and Data Locality in Hadoop.	L1,CO4,2M
6.	What is MapReduce?	L1,CO4,2M
7.	What is Apache Hive?	L1,CO4,2M
8.	Define DDL&DML?	L1,CO4,2M
9	Define HiveQL	L1,CO4,2M
10	Define Logical Joins,	L1,CO4,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain various data formats used in Hadoop. Discuss the advantages of analyzing Big Data using Hadoop and the concept of scaling out.	L2,CO4,10M
2	Describe the architecture of the Hadoop Distributed File System (HDFS). Explain its components and working with a neat diagram.	L2,CO4,10M
3	Write a short Note on fault tolerance, data replication, in HDFS. Discuss their significance in distributed computing.	L2,CO4,10M
4	Explain the MapReduce architecture. Describe the complete MapReduce process flow with a neat diagram.	L2,CO4,10M
5	Discuss the Java Interface, Data Flow, Hadoop I/O, Data Integrity, Compression, and Serialization in Hadoop.	L2,CO4,10M
6	Explain the architecture and features of Apache Hive. Discuss the data types and file formats supported by Hive.	L2,CO4,10M
7	Explain HiveQL. Discuss HiveQL Data Definition Language (DDL) and Data Manipulation Language (DML) commands with suitable examples.	L2,CO4,10M
8	Explain Logical Joins and Window Functions in Hive. Illustrate their usage with suitable examples.	L2,CO4,10M
9	Discuss various optimization techniques in Hive. Explain Table Partitioning, Bucketing, and Indexing with suitable examples.	L4,CO4,10M
10	Explain different Join Strategies in Hive. Discuss how query optimization techniques improve the performance of Hive queries..	L2,CO4,10M
11	Write a short Note on High Availability and Data Locality in HDFS. Discuss their significance in distributed computing.	

Unit-4

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define Apache Spark.	L1,CO5,,2M
2.	What are Lazy Evaluation and In-Memory Processing in Apache Spark?	L1,CO5,2M
3.	Define Directed Acyclic Graph (DAG)	L1,CO5,2M
4.	What is a Resilient Distributed Dataset (RDD)?	L1,CO5,2M
5.	Define Actions and DataFrames.	L1,CO5,2M
6.	Define Actions and DataFrames.	L1,CO5,2M
7.	What are Complex Data Types and JSON in Apache Spark?	L1,CO5,2M
8.	Define Grouping, Window Functions, and Joins in Spark SQL.	L1,CO5,2M
9	What are Data Sources, Broadcast Variables, and Accumulators in Apache Spark?	L1,CO5,2M
10	What are the different deployment modes of Apache Spark?	L1,CO5,2M

S.No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain the architecture and features of Apache Spark. Discuss the advantages of Apache Spark over Hadoop.	L2,CO5,10M
2	Explain the concepts of Lazy Evaluation and In-Memory Processing in Apache Spark. Discuss their significance in improving application performance.	L2,CO5,10M
3	Describe the Directed Acyclic Graph (DAG) architecture in Apache Spark. Explain the roles of SparkContext and SparkSession in Spark applications.	L2,CO5,10M
4	Explain the concept of Resilient Distributed Datasets (RDDs). Discuss Narrow Transformations, Wide Transformations, and Actions with suitable examples.	L2,CO5,10M
5	Explain DataFrames in Apache Spark. Discuss the process of converting RDDs into DataFrames and explain the role of the Catalyst Optimizer in query optimization.	L2,CO5,10M
6	Discuss various DataFrame Transformations in Apache Spark. Explain how Spark handles Dates, Timestamps, and Null values with suitable examples.	L2,CO5,10M
7	Explain the handling of Complex Data Types and JSON data in Apache Spark with suitable examples.	L4,CO5,10M
8	Discuss the concepts of Grouping, Window Functions, and Joins in Apache Spark SQL with suitable examples.	L2,CO5,10M
9	Explain the different Data Sources supported by Apache Spark. Discuss the role of Broadcast Variables and Accumulators in distributed	L2,CO5,10M

	computing.	
10	Discuss the deployment of Apache Spark in On-Premises Cluster environments. Explain the various Cluster Managers, including Standalone Mode and Spark on YARN.	L2,CO5,10M

Unit-5

S.No	2MarksQuestions(Short)	[BTLevel][CO][Marks]
1	Define Spark Performance Tuning	L1,CO6,2M
2	What is Stream Processing?	L1,CO6,2M
3	Define Event Time and Stateful Processing in Structured Streaming.	L1,CO6,2M
4	What is a Window operation?	L1,CO6,2M
5	What are Watermarks?	L1,CO6,2M
6	What is meant by dropping duplicates in a stream?	L1,CO6,2M
7	Define Structured Streaming. List its core concepts.	L1,CO6,2M
8	What are Transformations on Streams?	L1,CO6,2M
9	What are Input Sources and Output Sinks in Structured Streaming?	L1,CO6,2M
10	Mention any four real-world applications of Spark Structured Streaming.	L1,CO6,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain Spark Performance Tuning techniques. Discuss various methods to optimize the performance of Spark applications.	L4,C06,10M
2	Explain the fundamentals of Stream Processing. Discuss the architecture, characteristics, and applications of Stream Processing.	L4,C06,10M
3	Explain the concepts of Event Time and Stateful Processing in Structured Streaming with suitable examples.	L2,C06,10M
4	Discuss Window operations based on Event Time. Explain Tumbling Windows with suitable examples and applications.	L2,C06,10M
5	Explain the concept of Watermarks in Structured Streaming. Discuss how Spark handles late-arriving data and duplicate records in data streams.	L2,C06,10M
6	Explain the fundamentals of Structured Streaming. Discuss its architecture and core concepts with suitable examples.	L2,C06,10M
7	Discuss the working of Structured Streaming in Action. Explain the execution model and real-time processing workflow with suitable examples.	L4,C06,10M
8	Explain various Transformations on Streams in Structured Streaming with suitable examples.	L2,C06,10M
9	Discuss the different Input Sources and Output Sinks supported by Spark Structured Streaming with suitable examples.	L4,C06,10M
10	Explain the complete workflow of Spark Structured Streaming by discussing Stream Processing, Event-Time Processing, Window Operations, Transformations, and Input–Output mechanisms with suitable examples.	L2,C06,10M

Signature of the Staff:

Signature of Department Academic Committee Member 1:

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